Adaptive Horizon Scheduling in Monte Carlo Tree Search

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Problem Statement

Planning under uncertainty is a fundamental challenge in autonomous exploration, particularly in applications such as environmental monitoring and ocean sensing where efficient data collection is critical to field operations. Uncertainty arises in these scenarios when the robot's action effects are partially stochastic or the robot's sensors only collect partial, indirect information about the environment.

Monte Carlo Tree Search (MCTS), a simulation-based planning framework that incrementally builds a search tree by balancing exploration of uncertain actions and exploitation of high-reward trajectories through stochastic rollouts and value estimation (Flaspohler et al.), has been widely applied to informative path planning. However, its practical performance is highly sensitive to the design of the planning horizon, which, with the reward function, governs the balance between exploration for information gathering and exploitation for task performance. A fixed horizon often leads to suboptimal behavior: overly long horizons may result in inefficient exploration, while short horizons can cause premature exploitation and failure to identify globally informative regions.

Approach and Methodology

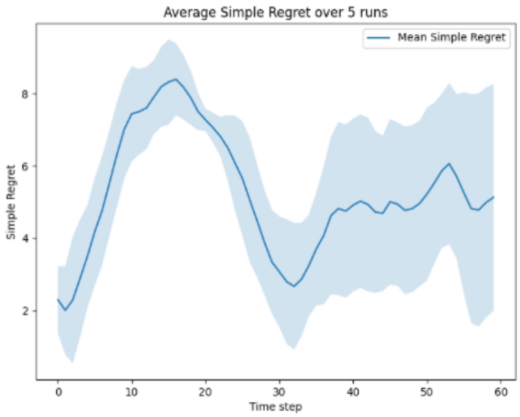
The proposed methodology is evaluated within a simulation environment designed for a mobile robot source-seeking scenario. In this setup, an autonomous agent must navigate an unknown spatial field to locate and characterize a signal source modeled by a Gaussian Process (GP), which provides a probabilistic map of the environment (Preston et al.). The evaluation schema focuses on the agent's ability to minimize uncertainty while maximizing information gain, using average simple regret, defined as the expected difference between the global maximum and the best-found observation (Bubeck et al.) as the primary metric to assess how closely the robot's sampled path approaches the global maximum of the source distribution.

This research initially evaluates several adaptive horizon scheduling strategies based on heuristics, including increasing, decreasing, and exponential growth modes. Although these mechanisms offer intuitive paradigms for horizon modulation, empirical analysis reveals significant structural instabilities. These baseline approaches are prone to oscillatory planning behavior and high stochastic variance. Furthermore, they frequently fail to converge reliably to globally informative regions, often becoming trapped in suboptimal horizons that do not align with the actual information density or evolving gradients of the source field.

To address these limitations, we propose a continuous adaptive horizon scheduling framework that integrates reward-based confidence and model uncertainty into both horizon selection and trajectory evaluation. Unlike discrete heuristic strategies, the proposed method enables a smooth transition between exploration and exploitation by dynamically scaling a spatial regularization term based on both temporal progression and uncertainty reduction.

This formulation encourages broad exploration in early stages while progressively biasing the planner toward high-reward regions as confidence increases. A temporally smoothed, confidence-driven mechanism further stabilizes the transition, reducing oscillatory behavior and improving convergence reliability.

Key Results



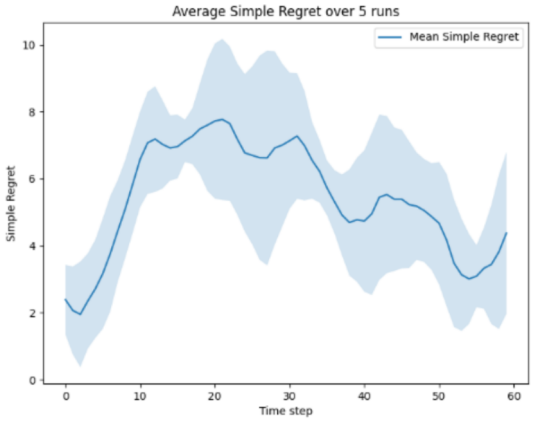


Figure 1: Comparison of baseline exponential scheduling (left) and the proposed adaptive framework (right).

Experimental results demonstrate that the proposed approach significantly improves stability and robustness compared to baseline adaptive strategies. Notably, the proposed approach exhibits a faster and more consistent decline in average simple regret after the initial exploration phase (approximately the first 10 timesteps), compared to baseline methods. While stochastic variance persists, the adaptive model demonstrates improved stability without the characteristic oscillations seen in baseline models. Variance across multiple runs remains bounded, indicating improved reliability for autonomous operations.

Relevance to the OCEANS Community

The project is highly relevant to the OCEANS community, where autonomous systems are increasingly deployed for seabed mapping, environmental monitoring, and adaptive sampling. In these domains, efficient use of limited sensing resources and robust decision-making under uncertainty are critical. By improving the stability of exploration–exploitation trade-offs, the proposed method enhances the reliability of autonomous planning in complex, uncertain environments.

References

[1] G. Flaspohler et al., ”Information-Guided Robotic Maximum Seek-And-Sample in Partially Observable Continuous Environments,” IEEE Robotics and Automation Letters, 2019.

[2] V. L. Preston et al., ”Phortex: Physically-Informed Operational Robotic Trajectories for Scientific Expeditions,” IEEE Transactions on Field Robotics, 2024.

[3] S. Bubeck et al., ”Pure Exploration for Multi-Armed Bandit Problems,” arXiv:0802.2655.